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OBJECT TRACKING METHOD AND SYSTEM USING CALIBRATED CAMERA

Abstract

An image taken of an object in motion is received. At least three markers on the object in motion in the image are detected. Image points of the three markers are determined. Based on the image points of the three markers, depths of the object in motion at the image points are determined, where the depths are determined relative to an image plane of the image. Using the image points and the depths, three-dimensional (3D) world points representing a position of the object in motion in a 3D real world coordinate are determined.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation-in-part application of U.S. patent application Ser. No. 18/802,634 filed Aug. 13, 2024, which is a divisional application of U.S. patent application Ser. No. 18/595,592, filed Mar. 5, 2024, the entire contents of which are incorporated by reference herein.

TECHNICAL FIELD
[0002] The present disclosure relates to a camera calibration method and a method and system of using a calibrated camera(s) for tracking and measuring a motion of a target object in a three-dimensional space.
BACKGROUND
[0003] Fundamentally, a camera provides an image mapping of a three-dimensional space onto a two-dimensional space or image plane. Current camera calibration techniques supply model parameter values that are needed to compute line of sight rays in space that corresponds to a point in the image plane.
[0004] A calibration or "projection" matrix, which is estimated during a camera calibration, is typically decomposed into eleven geometric parameters that define the standard pinhole camera model. Typically, camera model parameters include extrinsic and intrinsic parameters. The extrinsic camera parameters include 3D location and orientation of a camera in the world and intrinsic camera parameters include, among others, a focal length and relationships between pixel coordinates and camera coordinates.
[0005] In many applications, camera calibration is necessary to recover 3D quantitative measures about an observed scene from 2D images. For example, from a calibrated camera, it can be determined how far an object is from the camera, or the height of the object, etc. Typical calibration techniques use a 3D, 2D or 1D calibration object whose geometry in 3D space is known with very good precision.
[0006] From a set of world points and their image coordinates, one object of the camera calibration is to find a projection "matrix" and subsequently find intrinsic and extrinsic camera parameters from that matrix in a decomposition step. However, the decomposition into extrinsic and intrinsic camera parameters is one of the major issues in calibration due to reprojection error. Further, in the decomposition step, to extract the extrinsic and intrinsic camera parameters, several assumptions and constraints are made, which might be not true, for e.g., no lens distortion or no tilt.
[0007] Accordingly, it is desirable to have a camera calibration method that does not require decomposition into intrinsic and extrinsic camera parameters. In addition, it is also desirable to have a camera system for taking measurements that avoids having to make any assumptions regarding intrinsic or extrinsic camera parameters.
SUMMARY
[0008] There is provided a camera system and method for taking measurements of a moving object that avoids having to make any assumptions regarding intrinsic camera parameters.
[0009] Further, there is provided a camera system and method for taking measurements of a moving object that avoids having to split camera parameters apart from the camera's calibration matrix and as a result, the camera is ready to work with any lenses, any shift or tilt (intentionally or unintentionally) in the setup of the camera for tracking object in motion.
[0010] Additionally, there is provided a camera system calibration method for calibrating a camera used in taking measurements of a moving object without the decomposition of camera parameters into extrinsic and intrinsic parts.
[0011] In an embodiment, the camera system and method include a single camera device.
[0012] In one embodiment, during the calibration process, a virtual reference is aligned to a physical object in a global reference space to obtain the camera parameters.
[0013] A robust camera calibration system and method and system whereby a user can use any camera without the need for fine tuning the camera parameters to a global reference.
[0014] According to one aspect, there is provided a method for tracking an object in motion. The method comprises: capturing, from each of one or more calibrated cameras, one or more image frames of an

object in motion, each of the one or more calibrated cameras having been calibrated according to a calibration method that generates and uses a respective transformation matrix for mapping three-dimensional (3D) real world model features to corresponding two-dimensional (2D) image features; and determining, using a hardware processor, motion characteristics of the object in motion based on the captured one or more image frames from each of the one or more calibrated cameras, the determining of motion characteristics based on implicit intrinsic camera parameters and implicit extrinsic camera parameters of the respective transformation matrix from each respective one or more calibrated cameras.

[0015] In a further aspect, there is provided an object tracking system. The object tracking system includes a camera system comprising one or more calibrated cameras, each camera capturing one or more image frames of a position of an object in motion, each of the one or more calibrated cameras having been calibrated according to a calibration method that generates and uses a respective transformation matrix for mapping three-dimensional (3D) real world model features to corresponding two-dimensional (2D) image features; and a hardware processor coupled to a memory storing instructions that, when executed by the processor, configure the hardware processor to determine motion characteristics of the object based on the captured one or more image frames, the determining of motion characteristics of the object is based on implicit intrinsic camera parameters and implicit extrinsic camera parameters of the respective transformation matrix from each respective one or more calibrated cameras.

[0016] In accordance with a further aspect, there is provided a method of calibrating a camera. The method includes providing a transformation matrix (H) that represents a plurality of camera parameters; and aligning 2D image features (q) with 2D image features of a reference feature (q′) by applying one or more corrections (H.sub.Δ) to the plurality of the implicit camera parameters of the transformation matrix to obtain an updated transformation matrix (H′), thereby calibrating the camera, wherein the 2D image features (q) and 2D image features of the reference (q′) are represented in pixel coordinates.

[0017] Further to this aspect, the plurality of the camera parameters includes implicit extrinsic and implicit intrinsic camera parameters and further, the camera is calibrated without decomposing the plurality of the implicit camera parameters into explicit extrinsic and explicit intrinsic camera parameters.

[0018] Further to this aspect, the method of calibrating the camera includes modifying one or more implicit extrinsic camera parameters. The modifying comprises adjusting implicit extrinsic parameters through an affine correction. The affine correction recovers 6 Degrees of Freedom (DoF) representing the explicit extrinsic camera parameters.

[0019] In some embodiments, a system is provided that includes at least one memory device and at least one processor coupled with the memory device, where the at least one processor configured to at least the operations: receive an image taken of an object in motion; detect in the image, at least three markers on the object in motion; determine image points of the three markers; based on the image points of the three markers, determine depths of the object in motion at the image points, the depths determined relative to an image plane of the image; and using the image points and the depths, determine three-dimensional (3D) world points representing a position of the object in motion in a 3D real world coordinate.

[0020] In some embodiments, a method performed by a least one processor is provided. The method includes receiving an image taken of an object in motion. The method also includes detecting in the image, at least three markers on the object in motion. The method further includes determining image points of the three markers. The method also includes, based on the image points of the three markers, determining depths of the object in motion at the image points, the depths determined relative to an image plane of the image. The method also includes, using the image points and the depths, determining three-dimensional (3D) world points representing a position of the object in motion in a 3D real world coordinate.

[0021] A computer readable storage medium storing a program of instructions executable by a machine to perform one or more method described herein is also provided.

[0022] Further features as well as the structure and operation of various embodiments are described in detail below with reference to the accompanying drawings. In the drawings, like reference numbers indicate identical or functionally similar elements.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0023] FIG. **1** is an exemplary diagram of an object measurement apparatus employing a plurality of camera devices calibrated in accordance with methods of the present disclosure;

[0024] FIG. **2** depicts an application of the present disclosure to perform a mono camera calibration method without performing a decomposition to extract intrinsic and extrinsic camera parameters according to an embodiment;

[0025] FIG. **3** depicts an application of the present disclosure to perform a field camera calibration method, including the generating of an updated matrix H′ by providing an affine correction (extrinsic camera parameter);

[0026] FIG. **4** is an exemplary diagram illustrating a calibration method for the exemplary camera of FIG. **1** according to an embodiment;

[0027] FIG. **5**A depicts a method of conducting an on-site or field calibration of exemplary camera device in an example processing step of the calibration method according to an embodiment;

[0028] FIG. **5**B depicts a further method for field calibration/correction in an example processing step of the calibration method according to an embodiment;

[0029] FIG. **6** depicts an application of the present disclosure to perform a measurement (tracking parameters) of a moving object in a ‘hard-sync’ method using synchronous operating cameras;

[0030] FIG. **7** depicts an application of the present disclosure to perform a measurement (tracking parameters) of a moving object in a ‘soft-sync’ method using asynchronous operating cameras;

[0031] FIG. **8**A conceptually depicts an implicit stereo representation for a given trajectory model that assumes a soft sync, according to an embodiment; and

[0032] FIG. **8**B depicts an application of an inappropriate triangulation (“hard-sync” method) over “soft-sync” data according to an embodiment.

[0033] FIG. **9** shows an example of an object with three markers in some embodiments.

[0034] FIG. **10** shows an example of an image taken of an object in motion with three markers in some embodiments.

[0035] FIG. **11** shows another perspective of the image captured of an object in motion in some embodiments.

[0036] FIG. **12** is a flow diagram illustrating a method of finding position of an object in motion in some embodiments.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0037] Notably, technical terms used in the present specification are only for describing specific embodiments and are not intended to impose any limitation on the scope of the present disclosure. In addition, the term used in the present specification, although expressed in the singular, is construed to have a plural meaning, unless otherwise meant in context. The phrase “is configured with,” “include,” or the like, which is used in the present specification, should not be construed as being used to necessarily include all constituent elements or all steps that are described in the specification, and should be construed in such a manner that, among all the constituent elements or among all the steps, one or several constituent elements or one or several steps, respectively, may not be included, or that one or several other constituent elements, or one or several other steps may be further included.

[0038] In addition, in a case where it is determined that a detailed description of the well-known technology in the relevant art to which the present disclosure pertains makes indefinite the nature and gist of the technology disclosed in the present disclosure, the detail description thereof is omitted from the present specification.

[0039] As referred to herein, camera calibration is the process of obtaining camera parameters in the form of extrinsic and/or intrinsic parameters. Methods typically employed in the camera calibration include but are not limited to 3D Direct Linear Transformation (DLT), QR decomposition and perspective n point (PnP). A 3D DLT calibration obtains a homography or transformation matrix. The QR decomposition splits homography matrix into the extrinsic matrix and intrinsic matrix (of specific intrinsic model). The PnP such as P3P where n=3, obtains extrinsic camera parameters only, assuming intrinsic parameters are already known. Camera calibration method may use prior calibration results by the same or a different method. With prior information, camera parameters may be evaluated iteratively for e.g. using gradient descent (GD), from defaults to the ground truth.

[0040] As further referred to herein, DLT is one method for mono camera calibration, with its first step to recover the homography matrix H according to equation (1) as follows:

[00001] $H = \{h_{xx}, h_{xy}, h_{xz}; h_{yx}, h_{yy}, h_{yz}; h_{zx}, h_{zy}, h_{zz}; h_x, h_y, h_z\}$ (1)

from a calibration pattern, a set of feature points {right arrow over (p)}={p.sub.x,p.sub.y, p.sub.z} with known, predefined positions in 3D world, and from image points {right arrow over (q)}={q.sub.x,q.sub.y,1} as a solution of the following equations: {right arrow over (p)} H∝{right arrow over (q)} or {right arrow over (p)} H×{right arrow over (q)}=0 or according to equation (2) as:

[00002] $q_x = \frac{p_x h_{xx} + p_y h_{yx} + p_z h_{zx} + h_x}{p_x h_{xz} + p_y h_{yz} + p_z h_{zz} + h_z}; q_y = \frac{p_x h_{xy} + p_y h_{yy} + p_z h_{zy} + h_y}{p_x h_{xz} + p_y h_{yz} + p_z h_{zz} + h_z}$ (2)

To solve the system of equations above, this relation is rearranged into the form: Ah=0 where A is built by eq.1 from a set of 3D points {right arrow over (p.sub.n)} and a set of corresponding 2D image points {right arrow over (q.sub.n)}, and h={h.sub.xx;h.sub.yx;h.sub.zx;h.sub.x;h.sub.yx;h.sub.yy;h.sub.yz;h.sub.y;h.sub.zx;h.sub.zy;h.sub.zz;h.sub.z} is the vector of unknown H matrix elements. Then, a suitable optimization method can be applied, for e.g., least mean square (LMS), single value decomposition (SVD) or gradient descent (GD) and the system of equations is solved up to the scale, i.e. with 12−1=11 DoF.

[0041] As further referenced herein, PnP uses a set of methods for camera pose estimation, e.g., 6 DoF extrinsic parameters, assuming intrinsic parameters are given or known such as may be the case when the camera has been prior "calibrated". For this reason, PnP is not considered a complete calibration method since the intrinsic parameters are assumed. A minimum number of 3D points for camera pose estimation is three (e.g., P3P, three pairs of correspondences). As decomposition is not applied in the method for calibrating camera described in the current disclosure, PnP including P3P is not used in a factory calibration step. In some embodiments, the PnP or P3P is therefore used in a field calibration step only. That is affine corrections are applied to the default homography matrix, formally for the "uncalibrated" or "partially calibrated" camera, with intrinsic parameters unknown explicitly and hidden within the homography matrix.

[0042] As referred to herein, an "uncalibrated" or a "partially calibrated" camera is a camera for which pose estimation is either unknown or incomplete. In some embodiments, the pose estimation includes camera orientation RT and camera position −sR.sup.T. In some embodiments, a homography matrix without the decomposition may provide an incomplete pose estimation, for e.g. position only, but not the orientation. Camera orientation is typically defined by optical axis. However, the optical axis is an intrinsic camera parameter (such as principal points: c.sub.x, c.sub.y).

[0043] Camera position is the translation part of a transform T placing the camera from its own coordinate system into the world coordinate system. Extrinsic camera matrix C is the inverse of transform matrix T according to equation (3) as follows:

[00003] $T = C^{-1} = \{R; s\}^{-1} = \{R^T; -sR^T\}$ (3)

[0044] Based on the above equation, camera position p.sub.c obtained from extrinsic camera matrix C is: p.sub.c=−sR.sup.T. In some embodiments, the camera position p.sub.c may also be obtained from translation part from the inverse of the homography matrix H (without decomposition). Hence, the camera position p.sub.c=−sR.sup.T is considered to be independent from the intrinsic matrix K.

[0045] As described herein, decomposition of intrinsic and extrinsic camera parameters is the process of splitting the homography matrix H having 11 DoF into extrinsic C (6 DoF) and intrinsic K (5 DoF) parts: H={R;s}K.

[0046] The decomposition of camera parameters into extrinsic and intrinsic parts has an ambiguity as the decomposition gives rise to multiple possible solutions, for e.g., four—for QR decomposition for a given intrinsic model: H=C.sub.n K.sub.n.

[0047] As used herein, "QR decomposition" decomposes "rotational" part (RK) of the homography matrix H into an orthonormal rotation matrix and a lower (or upper) triangular.

[0048] Applying different intrinsic models Km may lead to even more distinctive solutions. One intrinsic model may be converted into any other models by applying rotation matrix R according to the following equation (4):

[00004] $H = C_1 K_1 = (C_1, R_{12})(R_{12}^T K_1) = C_2 K_2 \quad (4)$

where K.sub.1 is a first intrinsic model and K.sub.2 is a second intrinsic model, respectively with R.sub.12 is the rotation matrix for converting K.sub.1 to K.sub.2. By keeping the same original homography matrix H intact, distinct intrinsic models lead to distinct camera orientation R. In some embodiments, the camera position may remain the same p.sub.c=−SR.sup.T, irrespective of the intrinsic model K applied.

[0049] As described herein, the homography matrix H is a 4×3 matrix, defined up to an arbitrary scale. Thus, the homography matrix H has 11 DoF. Homography matrix H is the product of extrinsic matrix C and intrinsic matrix K as defined according to the following equation (5):

[00005] $H = \text{CK} \text{where} C = \{R; s\} \text{sothat} H = \{\text{RK}; \text{sK}\} \quad (5)$

[0050] The homography matrix H that maps 3D points {right arrow over (p)}={p.sub.x,p.sub.y,p.sub.z} to 2D image points {right arrow over (q)}={q.sub.x,q.sub.y,1} as: {right arrow over (p.sub.n)}H∝{right arrow over (q.sub.n)} (up to scale, proportional to) including projection into an image plane. In some embodiments, the homography matrix H is the result of the 3D DLT method. While DLT is one method, other suitable methods may be used including methods as proposed by: Richard Hartley & Andrew Zisserman, “Multiple View Geometry in Computer Vision”, page **88**.

[0051] As described herein, the extrinsic camera matrix C is a 4×3 matrix. Matrix C comprises a rotation matrix R and a translation vector s: C={R; s}. The extrinsic camera matrix C has six independent parameters (6 DoF) i.e. 3 DoF for the rotation and 3 DoF for the translation. The extrinsic camera matrix C maps 3D points from the world 3D coordinate system to the camera 3D coordinate system, without applying a projection onto the image plane.

[0052] The intrinsic camera matrix K is a 3×3 matrix. It comprises, among others, inner parameters of the camera such as: focal length, principal axis (shift), and tilts. The intrinsic camera matrix K may be represented according to equation (6) or (7) as:

[00006] $K = \{f_x, 0, 0; s, f_y, 0; c_x, c_y, 1\} (\text{“astigmatic - skew”model}) \quad (6)$ or

$K = \{f, 0, t_x; 0, f, t_y; c_x, c_y, 1\} (\text{“tilt - shiftmodel}) \quad (7)$

The intrinsic camera matrix K has five independent parameters (5 DoF).

[0053] A 3D rotation matrix “R”, a 3×3 orthonormal matrix, as used herein, is the component of a general 3D transform T or the extrinsic camera matrix C=T.sup.−1. Rotation matrix “R” holds 3 independent parameters (3 DoF). Other equivalent representations of 3D rotation include axis-angle and unit quaternion, all having 3 DoF.

[0054] A translation vector “s”, a 3D vector, as used herein, is the component of the general 3D transform T or the extrinsic camera matrix C=T.sup.−1. Translation vector “s” holds 3 independent parameters (3 DoF).

[0055] As referred to herein, Degrees of Freedom (DoF) refers to the number of independent parameters, possibly hidden. For example: rotation matrix, quaternion and the angle for axis-angle 3D rotation representation, each has 3 DoF.

[0056] As referred to herein and described in the present disclosure, camera calibration is the process of obtaining camera parameters in the form of extrinsic and/or intrinsic parameters. In some embodiments, the camera calibration comprises a multiple steps process as will be described below. In some embodiments, the camera calibration comprises 2-steps calibration with a first step being referred to as a factory calibration and second step being referred to as a field calibration. In accordance with aspects of the present disclosure, the factory calibration is typically applied at least once for the camera to obtain a transformation matrix H representing a plurality of camera parameters. The plurality of camera parameters comprises intrinsic camera parameters and extrinsic camera parameters. In some embodiments, the transformation matrix H is for mapping 3D real world model features to corresponding 2D image features. In some embodiments, the transformation matrix is a homography matrix H. In some embodiments, the transformation matrix H represents intrinsic and extrinsic camera parameters. In some embodiments, the transformation matrix represents implicit intrinsic and implicit extrinsic camera parameters. In some embodiments, the transformation matrix may be obtained by DLT method as will be described below or other suitable algorithms. In some embodiments, the field calibration step may be applied once or multiple times. In some embodiments, the field calibration may be referred to as a calibration correction for the extrinsic camera parameters. In some embodiments, the second step is for correcting the extrinsic camera parameters implicitly, within the homography matrix H, without the decomposition of the intrinsic and extrinsic camera parameters. The field calibration may employ PnP algorithm including P3P, using the homography matrix H obtained from the first step or factory calibration as an initial estimation. Therefore, in the field calibration, only affine transform corrections are applied without altering intrinsic camera parameters defined implicitly within the homography matrix H.

[0057] FIG. **1** is an exemplary diagram of an object measurement apparatus **10** employing a plurality of camera devices **100**.sub.1, **100**.sub.2, . . . , **100**.sub.n calibrated in accordance with methods of the present disclosure. Each of the plurality of camera devices are capable of wired or wireless communication with a computing device **200** for performing object tracking and parameter measuring operations.

[0058] Each camera device, such as the exemplary camera device **100** shown in FIG. **1**, according to the embodiment of the present disclosure, may include at least components such as: a lens **101** and sensor **105**, a digital signal processor or controller **110**, which can include one or more hardware microprocessors, a communication unit **120** that is connected to and controlled by the controller **110**, an image capture unit **130** for capturing images and/or video, a calibration unit **140** for controlling camera calibration operations, a memory **150**, and a tracking unit **160**.

[0059] Exemplary camera device **100** as depicted in FIG. **1**, in connection with embodiments of the present disclosure, may include any device, system, component, or collection of components configured to capture images and/or video. In an embodiment, the camera device **100** includes an image capture unit **130** having optical elements such as, for example, lenses **101**, filters, etc., and at least one image sensor **105** upon which an image may be recorded. Such an image sensor **105** may include any device that converts an image represented by incident light into an electronic signal. The image sensor **105** may include a plurality of pixel elements, which may be arranged in a pixel array (e.g., a grid of pixel elements) for acquiring an Image. For example, the image sensor **105** may comprise a charge-coupled device (CCD) or complementary metal-oxide-semiconductor (CMOS) image sensor. The pixel array may include a two-dimensional array with any aspect ratio. The image sensor **105** may be optically aligned with various optical elements that focus light onto the pixel array, for example, lens **101**. Any number of pixels may be included such as, for example, hundreds or thousands of megapixels, etc.

[0060] In an embodiment, the image capture unit **130** acquires an image under the control of the controller **110**, and the acquired image is stored in the memory storage unit **150** under the control of the controller **110**. Further data and information for operation of the camera **100** can be stored in the memory **150**. Then, the image acquired in the image capture unit **130** is stored in the memory **150**. The acquired images have respective pieces of unique identification information, and the pieces of identification information can correspond to points of time, respectively, at which the images are acquired. In addition, other types of tracking information that are detected from at least one image according to the target designation information can be stored in the memory **150**. A storage area of the memory **150**, in which the images are stored, can be hereinafter alternatively referred to as an image storage unit **152**, and a storage area of the memory **150**, in which “tracking information” associated with target object tracking are stored, is hereinafter referred to as a tracking information storage unit **155**. The camera memory **150** additionally stores local camera calibration information such as the value of extrinsic and intrinsic parameters used for camera calibration. In some embodiments, the camera memory **150** may also store the value of extrinsic and intrinsic camera parameters obtained from the camera calibration.

[0061] An image processing unit **170** is further provided to perform types of digital signal processing to raw camera image data, e.g., filtering, noise reduction, image sharpening, etc.

[0062] As further shown in FIG. **1**, exemplary camera **100** can include a communication unit **120** to enable the camera to receive remote control signals for performing camera operations in connection with object tracking and measuring movement parameters and includes at least one module for performing wired and/or wireless communication with a remote controller processor or remote computing apparatus

200 (remote control unit). The communication unit **120** receives a control signal from a remote control apparatus through at least such one module and can transfer an acquired image to the remote controller processor or computing apparatus **200** under the control of the controller **110**.

[0063] Whether remotely or locally controlled using controller **110**, the camera **100** may operate at certain frame rates or be able to capture a certain number of images in a given time. The camera **100** may operate at a frame rate of greater than or equal to about sixty frames per second (fps), for example between about one hundred and about three hundred frames per second (fps). In some embodiments, a smaller subset of the available pixels in the pixel array may be utilized to allow for the camera to operate at a higher frame rate.

[0064] The tracking unit **160** operates to detect a target that corresponds to a received target object being tracked. In some embodiments, the target comprises a ball used in golf, cricket, baseball, softball, tennis, etc.

[0065] In an embodiment, any number of a variety of triggers may be used to cause the exemplary camera **100** to capture one or more images of the moving object. By way of non-limiting examples, the camera **100** may be triggered upon receipt of a timed activation signal, or when the moving object is detected, known or estimated to be in the field of view of the camera, or when the moving object first begins or modifies its flight (for example when a baseball is pitched, when a baseball is batted, when a golf ball is struck, when a tennis ball is served, when a cricket ball is bowled, etc.), or when a moving object is detected at a leading row of pixels in the pixel array, etc.

[0066] In an embodiment, the remote controller processor or computing apparatus **200** can control overall operation of the remote cameras **100**, **100**.sub.1, **100**.sub.2, . . . , **100**.sub.n that can be physically spaced apart to perform object tracking and measurements according to the embodiment of the present disclosure. The computing apparatus **200** can control the image capture unit **130** according to the control signals **175** that is received through the communication unit **120** and to configure the controller **110** to acquire an image. Subsequently, the controller **110** can store the acquired image at the device and/or can transmit the stored image and the related image identification and tracking information to the computing apparatus **200** for further processing. The further processing can include the application of one or more algorithms to receive raw camera image data in a series of image frames. For example, in accordance with embodiments herein, an example camera measurement of a moving object may include, but is not limited to identifying, in each of the images, one or more of: a radius of the object, a center of the object, a velocity, an elevation angle, and an azimuth angle, e.g., calculated based on the radius of the object, the center of the object, and any pre-measured camera alignment values.

[0067] In accordance with embodiments herein, FIG. **2** depicts implementation of the method described in the present disclosure using calibration unit **140** to perform a factory camera calibration method **250** i.e. first step of camera calibration. In some embodiments, the factory calibration **250** is performed using given known 3D features and detected 2D images features from the camera thereby generating the transformation matrix H. Such transformation matrix may be termed as “default” transformation matrix. As described herein, the transformation or projection matrix H represents intrinsic and extrinsic camera parameters having 11 DoF. As can be seen from FIG. **2**, the factory calibration is undertaken without performing a decomposition to extract intrinsic and extrinsic camera parameters explicitly. In some embodiments, following the factory calibration, the camera may undergo a further calibration i.e. field calibration prior to using the calibrated camera. Hence, in some embodiments, the camera that has been calibrated following the factory calibration may be still considered as an “uncalibrated” camera. In some embodiments, the camera that has been calibrated following the factory calibration refers to a “partially calibrated” camera.

[0068] In accordance with further embodiments herein, FIG. **3** depicts an application of the present disclosure to perform a field calibration method **300**. In some embodiments, the field calibration refers to a second step of the camera calibration described in the present disclosure. In some embodiments, as described above, the field calibration refers to a method **300** for calibrating an ‘uncalibrated’ camera or a “partially calibrated” camera. In some embodiments, the field calibration is performed based on given known 3D features and detected 2D images features from the camera, and given default camera parameters from transformation matrix H, where the transformation matrix H is obtained from the first step of calibration i.e., factory calibration (refer to FIG. **2**). In some embodiments, the calibration unit

140 performs a method to generate an updated correction matrix H′ by performing affine correction to obtain updated extrinsic camera parameters. Thus, it is the case that the calibration method comprising the factory calibration and field calibration described herein advantageously is performed without decomposition of the intrinsic and extrinsic camera parameters. Accordingly, the accuracy of the calibration method described herein is advantageously better than the decomposition method as no assumption was made.

[0069] In some embodiments, there is provided a method for tracking an object in motion. The tracking method includes capturing, from each of one or more calibrated cameras, one or more image frames of the object in motion. In some embodiments, each of the one or more calibrated cameras may have been calibrated according to a calibration method described herein. In some embodiments, the calibration generates and uses a respective transformation matrix for mapping 3D real world model features to corresponding 2D image features. The tracking method further comprises determining, using a hardware processor, motion characteristics of the object in motion based on the captured one or more image frames from each of the one or more calibrated cameras. In some embodiments, the determining of motion characteristics is based on implicit intrinsic camera parameters and implicit extrinsic camera parameters of the respective transformation matrix from each respective one or more calibrated cameras. In some embodiments the tracking is a 3D tracking.

[0070] In some embodiments, each of the one or more calibrated cameras is calibrated without decomposing the plurality of the camera parameters into explicit extrinsic and explicit intrinsic camera parameters. In some embodiments, each of the first calibration (factory calibration) and the second calibration (field calibration) is performed without the decomposition of the camera parameters into explicit extrinsic and explicit intrinsic camera parameters. In some embodiments, both the first calibration step i.e. factory calibration and the second step calibration i.e. field calibration are performed without the decomposition of the camera parameters into extrinsic and intrinsic camera parameters.

[0071] In some embodiments, following the first step or factory calibration, each of the one or more calibrated cameras is further calibrated by modifying one or more implicit extrinsic camera parameters to obtain the explicit extrinsic camera parameters. In some embodiments, the modifying step comprises adjusting implicit extrinsic camera parameters through an affine correction.

[0072] FIG. **4** is an exemplary diagram illustrating the field calibration method **400** for the exemplary camera **100** of FIG. **1** according to an embodiment. In some embodiments, the field calibration method for e.g. recovery of camera position in 3D world by reference patterns (natural or artificial) is accomplished without using extrinsic and intrinsic camera parameters explicitly. That is, the field calibration method is performed without decomposition of camera parameters into intrinsic and extrinsic parts. The field calibration method may be performed by processing logic that may include hardware (processor, calibration unit **140**, circuitry, dedicated logic, etc.) and software (such as is run on the hardware processor **110** or computer system **200**), or a combination of both, which processing logic may be included in the calibration unit **140**.

[0073] Referring to FIG. **4**, in an initial step **403**, there is obtained the camera devices default or “given” camera parameters. In some embodiments, the default camera parameters may be inherent in an initial calibration, projection or “transformation” matrix (“H0”) that represents the plurality of default intrinsic and extrinsic camera parameters. In some embodiments, the camera is operated to obtain real world 3D feature points of an image obtained through the camera field of view (FoV) and are represented as feature points “p”. Default 2D feature points of the camera image plane are represented as q.sub.0, and desired 2D feature points are referred to as q.sub.K.

[0074] In some embodiments, the 3D models or reference patterns may include but are not limited to stencils. In some embodiments, the 3D models or reference patterns may be any suitable reference including cage corners, stumps, marks, lines, boundaries. In some embodiments, the 3D models or reference patterns may be static, i.e., its position does not change. In some embodiments, the 3D models or reference patterns may be in motion. In such an embodiment, a first camera system (for e.g. a stereo system, where there are at least two cameras) is calibrated using the method described herein and is used to track a moving object to obtain the 3D position and/or trajectory parameters. This information may be used to perform a calibration of a second camera system, where the second camera system may be the same or different stereo system as the first camera system. In such an embodiment, the second camera system is also used to track the same moving object to obtain the camera parameters of the second camera system.

[0075] There is also provided a step of controlling the camera to update the camera parameters with a 3D space, which is represented as an updated transformation matrix H.sub.k+1 based on the natural or artificial reference patterns described at step **406**. This step includes updating of the extrinsic camera parameters having 6 DoF. At step **406**, the camera with known or given intrinsic parameters (from the factory calibration) is manually calibrated by adjusting the camera parameters associated with a small axis-angle and/or a translation vector. In some embodiments, the adjustment impacts implicit extrinsic camera parameters having 6 DoF. Next, a manual aligning of obtaining 2D virtual image features (q2) may be performed with 2D image features of a reference image (q1) by applying one or more corrections matrix (H.sub.Δ) to the plurality of the implicit camera parameters of the transformation matrix H.sub.k, thereby obtaining the updated transformation matrix H.sub.k+1 and in turn calibrating the camera. As described above, the camera calibration is performed without the decomposition of the camera parameters into explicit intrinsic and extrinsic parts. In this embodiment, the 2D virtual image features (q2) and 2D image features of the reference image (q1) are represented in pixel coordinates.

[0076] Continuing at step **410** of FIG. **4**, based on the manual aligning as performed during the calibration at step **406**, the processor may build at the camera, a correction matrix, “H.sub.Δ”. Subsequently at step **414**, the built correction matrix H.sub.Δ is applied at step **410** to the initial transformation matrix H.sub.k associated with the default camera parameters, to obtain an updated matrix H.sub.k+1 associated with updated camera parameters according to the following equation (8):

[00007] $H_{k+1} = H \quad H_k \quad (8)$

[0077] Further, according to step **417** of FIG. **4**, once the updated matrix H.sub.k+1 is obtained at step **414**, the updated matrix H.sub.k+1 is subsequently applied to 3D models, where new 2D points may be obtained according to the following: q.sub.k+1∝pH.sub.k+1, where new 2D points (camera position) are obtained from the updated homography matrix H.sub.k+1. As 3D points p are static, 2D points q.sub.k are changing according to H.sub.k and converging to desired q.sub.K by tuning H.sub.k+1 with H.sub.Δ.

[0078] Continuing to step **420** of FIG. **4**, a determination is made as to whether the 2D image features (q) and 2D image features points of the virtual reference image (q′) are equal or below some target threshold number of pixels. In other words, it is determined whether the difference between a 2D position (q) evaluated using the current transformation matrix H.sub.k and a desired 2D position (q′) obtained from updated correction matrix H.sub.k+1, i.e., the quantity |q−q′| is less than or more than a threshold value “threshold”. In an embodiment, the threshold value is represented by a distance in pixels, e.g., one (1) pixel and is alternatively referred to as a “reprojection error”.

[0079] If, at step **420**, it is determined that the quantity |q−q′| is greater than a predefined threshold value, for e.g., greater than one pixel, then the method **400** repeats by returning to step **406** and all steps **406**, **410**, **414**, **417**, and **420** are repeated. In some embodiments, the steps of method **400** are iterative and repeated until such time when the quantity |q−q′| is equal to or less than the threshold value, for e.g., ≤1 pixel, at which time the method **400** terminates as these 2D points are considered “in place”. Following this, the camera is deemed to have been fully calibrated and is ready to be used for measurement purposes. The 2D feature points {right arrow over (q.sub.n)}={q.sub.nx,q.sub.ny,1} are computed using the obtained correction transformation matrix H.sub.k+1 and the set of feature points {right arrow over (p.sub.n)}={p.sub.nx,p.sub.ny,p.sub.nz} in 3D (Cartesian) coordinate space according to eq.2. The computed {right arrow over (q.sub.n)}={q.sub.nx,q.sub.ny,1} approximately yield the 2D position.

[0080] In an embodiment, an automatic camera calibration correction i.e. second stage calibration may be performed. The calibration computation method is described as follows. Solution for correction matrix H.sub.Δ={Δr.sub.k;Δs.sub.k} is determined from a system of linear equations:

[00008] $(p \quad r_k + \quad s_k)r_k \times q_K = \text{-}(\mathrm{pr}_k + s_k) \times q_K \quad (9)$

where equation (9) is derived from projection constraint described in equation (10)

[00009] $q_K \propto pH_{k+1} = pH \quad H_k = (\mathrm{pr}_{\;,k} + \quad s_k)r_k + s_k \quad (10)$

An angle approximation: r.sub.Δ,k≈(I+Δr.sub.k) is applied to generate a system of linear equations (approximate), where “I” is an identity matrix i.e. unit 3×3 matrix. A valid (orthonormal) rotation matrix

r.sub.Δ,k can then be constructed from obtained Δr.sub.k by applying Cayley formula and then apply H.sub.Δ={Δr.sub.k;Δs.sub.k} to obtain updated 2D points q.sub.k+1∝pH.sub.k+1 (using eq. (2). This procedure is repeated till the value of |q.sub.k+1−q.sub.K| is within the threshold.
r.sub.Δ,k˜I+Δr.sub.k, where Δr.sub.k is a skew symmetric matrix. Then, setting k=0, where k is an iteration counter, then the H.sub.k is initially provided by the following equation (11):
[00010] $H_k = H, H_k \equiv \{r_k, s_k\}.$ (11)
[0081] In some embodiments, the method subsequently performs the following steps: [0082] 1. solving (p Δr.sub.k+s.sub.Δ,k).Math.(q×r.sub.k)=(q×(pr.sub.k+s.sub.k) for Δr.sub.k and s.sub.Δ,k where r.sub.k represents a change in rotation and s.sub.k represents a change in camera translation; [0083] 2. assigning H′=H.sub.k if Δr.sub.k and s.sub.Δ,k are small and then stop; [0084] 3. otherwise, building a valid r.sub.Δ,k from Δr.sub.k (e.g., using half angles, Cayley formula in the event that the value does not converge. This is to avoid the use of polynomial equations); and [0085] 4. updating H.sub.k+1={r.sub.k+1=r.sub.Δ,k,r.sub.k,s.sub.k+1=s.sub.Δ,kr.sub.k+s.sub.k}, k=k+1 and repeat.
[0086] In an embodiment, the method solves a system of linear equations where the relation is between 3D points in a scene and their projection onto a camera's 2D the image plane such that the method solves a set of variables using the DLT method from the set of similarity relations: q∝p H′ where q (2D points) and p (3D points) are known vectors, where “x” denotes proportional to, and H′ is a matrix (or linear transformation) which contains the unknowns to be solved. According to this embodiment, a 2D camera position consisting of q.sub.x and q.sub.y from the transformation matrix H′ is obtained.
[0087] FIG. **5**A depicts an exemplary environment **550** in which method according to one embodiment of the present disclosure is applied for conducting an on-site calibration (field calibration) of exemplary camera device **100** in an example second stage of calibration process. The view shown in FIG. **5**A represents the image at a camera viewer. As shown in FIG. **5**A, the camera calibration unit configures the camera to generate a virtual reference image **555** which, for camera calibration at step **406** of FIG. **4**, is controllable for translational and rotational movement to align with a corresponding physical real world object **565** in the camera view for an affine correction. In an embodiment, the camera includes a physical directional button(s) or a manipulable “joystick” **575**, which is manipulated by a user to move the virtual reference image **555** such that the virtual stumps **555** match the physical stumps **565**, i.e., are aligned within a threshold distance (q-q′). The movement/alignment data obtained from the aligning of the calibration reference image **555** to the physical real world object **565** in the camera field of view is an offset correction that is used to generate new/updated extrinsic parameters of the camera.
[0088] In an embodiment, the method described in FIG. **4** recovers camera position in 3D world by reference patterns **565**, i.e., a stump or a wicket, without using extrinsic and intrinsic camera parameters explicitly, i.e., without decomposition of camera parameters into intrinsic and extrinsic parts. In an embodiment, natural or artificial reference patterns can be used to perform the affine correction. Artificial reference patterns **565** can include other types of physical calibration objects, e.g., a marker, corner of a cage and so forth.
[0089] FIG. **5**B depicts a further method for field calibration/correction. In an example sports cage **580** shown in FIG. **5**B, the camera viewer presents virtual reference image as marks **556**, e.g., marking a geometric shape such as a square, which can be a reference used for calibration. Other example of virtual references can include markings for corners of a cage (e.g., a batting cage), lines indicated as field line markings **557**, field markings, and so forth. The camera includes physical directional button(s) **592** or a manipulable “joystick” (not shown) controllable for adjusting the translational (e.g., offset) and rotational (e.g., angle) parameters **590** associated with the movement of the reference markings **556**, **557** when performing a calibration. The movement of the virtual reference calibration markings **556**, **557** in the camera view align with a corresponding physical real world object(s) in 3D space represented, e.g., by orthogonal x, y, z directions as in a Cartesian coordinate system, for an affine correction thereby impacting 6 DoF and updated extrinsic camera parameters are thus obtained.
[0090] Additionally, shown in the camera view of FIG. **5**B, homographic matrix values **595** for default (H.sub.n), correction (H.sub.Δ) and updated matrices (H.sub.n+1) for each calibration. From the updated homographic matrix values **595**, the camera position in 3D world is recovered without using intrinsic and extrinsic camera parameters explicitly, i.e., without decomposition of camera parameters into intrinsic

and extrinsic parts.
[0091] Other examples of reference markings include checkerboards, AprilTag, etc. and the calibration apparatus described in pending U.S. patent application Ser. No. 18/527,245 filed on Dec. 2, 2023, as well as a moving object including a ball. Natural patterns are known objects with a-priori placement with respect to camera. That is, in the first stage calibration, the relative position of the patterns to camera is known (a-priori). The parameters of the camera are recovered by the DLT method as described above, without decomposition of the camera parameters into intrinsic and extrinsic parts.
[0092] Thus, referring to the method **400** of FIG. **4**, there is provided a two-stage calibration process, wherein a first stage is the recovery of implicit camera parameters using the DLT to recover 11 DoF parameters (e.g., implicit 6 DoF extrinsic and 5 DoF intrinsic). Then, at step **410**, the second stage of the calibration requires an on-site affine correction, without altering implicit intrinsic parameters. This on-site correction assumes the following: the implicit intrinsic parameters of the camera do not change, i.e., the focal length, distortion profile, etc. remain the same; and only translation and/or rotation of camera with respect to world coordinate has changed. Thus, the extrinsic 6 DoF parameters are changed implicitly, without decomposing of combined set of calibration parameters (11 DoF) into intrinsic (5 DoF) and extrinsic parts (6 DoF).
[0093] In an embodiment, to extend a depth of focus along a field, e.g., of a stadium, the system further provides support for tilted lenses by performing a lens tilt calibration method such as that described in Scheimpflug principle by an implicit, unspecified model, of intrinsic camera parameters.
Measurements
[0094] The exemplary camera system of FIG. **1** is further configured as an object measurement system applicable to recover 3D world object positions by using stereo (two or more) cameras without decomposition of camera parameters into intrinsic and extrinsic parts. That is, according to embodiments herein, a measurement system and method for tracking objects is provided. In an embodiment, object tracking measurements include a hard-synchronization, and soft-synchronization parameter processing to recover projectile trajectories by a regularized high order trajectory model by partially overlapping or non-overlapping by field of view (FOV) stereo cameras for e.g. two or more without decomposition of camera parameters into intrinsic and extrinsic parts. It is to be understood that each of the two or more cameras in the object measurement system may have been calibrated according to the method described herein.
Hard Sync
[0095] For hard-sync object tracking and measurement, the camera system of FIG. **1** is configured such that the cameras, e.g., one or more multiple cameras **100**.sub.1, **100**.sub.2, . . . , **100**.sub.n use the same external VSync signal, with cameras hard wired by a general purpose I/O (GPIO) line, and where frames are captured synchronously with the same timestamp.
[0096] In some embodiments, FIG. **6** depicts an application to perform a measurement for e.g., tracking parameters of a moving object in a 'hard-sync' method **600**. In some embodiments, the hard-sync method uses synchronous cameras that have been calibrated according to the method described herein. In some embodiments, the cameras have been calibrated without decomposition of the camera parameters into intrinsic and extrinsic camera parameters during the calibration. In some embodiments, the hard-sync method further utilizes a 2D-to-3D conversion algorithm. Given parameters from N cameras (1, 2, . . . , N), providing respective projection matrices H.sub.n where n=1, 2, ... N, there are obtained corresponding tracked object features in 2D with each camera providing a 2D feature (t, q1), (t, q2), . . . , (t, qN). Then, applying a multi 2D-to-3D DLT, 3D positions of the moving object may be obtained with estimated errors.
[0097] In an embodiment, since pointwise triangulation methods are not used and are replaced with implicit "overlapping" of the trajectory in a parametric space, the requirement for "hard sync" can thus be relaxed.
[0098] As an example, given two cameras (camera "a" and camera "b") that have been calibrated according to the 2-stages calibration method described in the current disclosure, there are computed 2D feature points q.sub.ax and q.sub.ay for camera "a" and 2D feature points q.sub.bx and q.sub.by for camera "b" using the obtained correction transformation matrix H′. Subsequently, these 2D image points may then be converted to corresponding 3D real world points. The measurement to obtain 3D points

from 2D images is thus expedited as the measurement using the calibrated camera disclosed herein does not require a split of extrinsic and intrinsic parameters and the processing is not concerned with focal lengths, principal axes, lens tilts, stereo base distance.

Soft Sync

[0099] In accordance with a further embodiment herein, FIG. **7** depicts an application of the present disclosure to perform a measurement for e.g., tracking parameters of a moving object in a ‘soft-sync’ method **700**. In some embodiments, the soft-sync method uses asynchronous cameras that have been calibrated according to the method described herein. In some embodiments, the cameras have been calibrated without decomposition of the camera parameters into intrinsic and extrinsic camera parameters. In some embodiments, the soft-sync method utilizes non-overlapping FOV cameras. In some embodiments, the soft-sync method further utilizes a 2D-to-3D conversion algorithm. Given parameters from N cameras (1, 2, . . . , N), providing respective projection matrices H.sub.n where n=1, 2, ... N, there are obtained corresponding tracked object features in 2D with each camera providing a 2D feature, e.g., {t[1,n],×X[1,n],×Y[1,n]}) from camera 1, {t[2,n],×X[2,n],×Y[2,n]} from camera 2, . . . {t[m,n],×X[m,n],×Y[m,n]}) from camera m. Then, applying a 12p (twelve point) trajectory model, there are computed trajectory parameters with estimated errors from which 3D positions of the moving object are obtained with the corresponding estimated errors.

[0100] For soft-sync object tracking and measurement, the camera system of FIG. **1** is configured such that the cameras, e.g., one or more multiple cameras **100**.sub.1, **100**.sub.2, . . . , **100**.sub.n capture image frames asynchronously. The processing, however, uses the same common time reference by clock. In such an embodiment, each camera is synchronized periodically by precision time protocol (PTP) and/or global position system (GPS). Each camera captures frames in a free run mode with its own interval VSync, thus frames' timestamps are different, but the reference time is the same within a tolerance defined or required by the measurement precision. In some embodiments, for a ball speed of 50 m/s and a distance measurement of 2.5 cm, the time synchronization tolerance is about 0.5 milliseconds (ms). As an exemplary embodiment, for the time tolerance of 0.5 ms and clock oscillators with an accuracy of 10 points per million (ppm), the cameras are to be synchronized every 50 seconds.

[0101] FIG. **8**A conceptually depicts an implicit stereo representation **800** for a given trajectory model that assumes a soft sync. In this embodiment, there are recovered multiple cameras' asynchronous (“soft sync”) ball trajectory parameters by trajectory stitching in parametric space. As shown in FIG. **8**A, parameters are recovered from cameras **100**A and **100**B such that the trajectory is scaled until it stitches by shape and by time. The scaling is shown to obtain a trajectory and measurement of an object (e.g., 3D positions of a ball by time) along path **805**. However, “soft-sync” processing obtains trajectory parameters with camera **100**A frames along path **805**A only and with camera **100***g* frames along path **805**B. In some embodiments, paths **805**A and **805**B may overlap. In some embodiments, paths **805**A and **805**B may not overlap by cameras' view.

[0102] FIG. **8**B depicts inappropriate application of the triangulation (“hard-sync” method) over “soft-sync” data resulting in object's positions errors. As shown in FIG. **8**B, frames are captured asynchronously by a free running camera (using an internal VSync signal), using the same time reference (PTP, GPS, power grid), but without exact matching of timestamps. The method chooses the closest frames by pairs to provide a ground truth object location(s), e.g., ball positions **810**. The method estimates ball locations, e.g., a first location **815**A based on results of an applied triangulation over the two close frames, e.g., taken by cameras **100**A, **100**B or a second location **815**B based on the triangulation. In an example implementation, there is chosen the closest camera image frames, wherein for 250 fps (two milliseconds) and ball speed of 50 m/s (or 200 fps and ball speed of 40 m/s), in which it results in object's position estimation error (“a shift”).

[0103] This soft-sync method includes a feature such as: a 9p or 12p trajectory model, possibly non-overlapping views, e.g., using two, three or more cameras. Such an object tracking method can additionally be referred to as: non-overlapping stereo, multiple camera stereo, (implicit) trajectory stitching. This soft-sync method may be applied in cricket, baseball, football, etc.

[0104] In an embodiment, a “Triangulation” is applied to provide a 3D reconstruction for asynchronous (soft-sync) cameras with the common time reference and the synchronous frames captured by common VSync signal.

[0105] The trajectory is physically the same for all cameras. In this embodiment, it is further assumed that: all cameras share a common time reference, to provide proper timestamps even if their frames are asynchronous.
[0106] In an embodiment, a requirement for cameras' overlapping views can be relaxed by applying an “overlapping” in a parametric space. Two cameras might look in opposite directions and observe different part of the same trajectory. In some embodiments, sufficient base distance between cameras may be necessary.
[0107] A requirement for cameras to have a close focal length can therefore be relaxed. In some embodiments, the cameras may have significantly different focal length with small or no FOV overlapping.
No Sync
[0108] For no-sync, object tracking and measurement, the camera system of FIG. **1** is configured such that the cameras, e.g., one or more multiple cameras **100**.sub.1, **100**.sub.2, . . . , **100**.sub.n run asynchronously, using their own time reference by their own clock. In some embodiments, for no-sync, clock synchronization may be recovered (i.e. conversion from “no-sync” to “soft-sync”) by post-processing using common events, by for e.g. video, audio, vibration or radar. Video events may be but are not limited to ball bouncing and IR flash. Audio and/or vibration events may be but not limited to ball hit and bouncing sound. Radar events may be but are not limited to ball release and/or ball bouncing. In some embodiments, clock synchronization may be recovered by flying ball. In such an embodiment, a specific configuration of cameras positions may be required for e.g. “in-plane” configuration where cameras are positioned side-by-side.
[0109] In an exemplary embodiment, for a camera system having two cameras i.e. camera A and camera B, the camera system is configured to determine the position of the object in motion. Camera A is configured to capture first and third images at first and third time points, respectively. Camera B is configured to capture second image at second time point. The third time point is after the first time point and the first time point is after the second time point. In another exemplary embodiment, the second time point is after the third time point and the third time point is after the first time point. Yet, in another exemplary embodiment, the third time point is after the second time point and the second time point is after the first time point. In some embodiments, cameras A and B are positioned at different locations.
[0110] In some embodiments, a system is provided that finds position and/or direction (e.g., orientation) of an object in motion, including a ball striking element such as a baseball bat, softball bat, cricket bat, golf club, or another object in motion. In some embodiments, finding the position and/or direction of the object in motion includes finding the position and/or direction of the ball striking element when a player or a user swings the ball striking element during practice or the actual game. The object has at least three markers. In some embodiments, as a camera, for example, a single or mono camera, captures images of the object in motion (e.g., frame by frame), image points of the three markers on a frame are recovered. The system, based on the recovered image points and using a homography matrix H and cross-ratio, determines the depths of the object at the image points, the depth being relative to the image plane of the image. Based on determined depth, the system calculates three-dimensional (3D) world points as positions. In some embodiments, the three markers are distinct, and therefore, the system can also determine the direction or orientation of the object based on the determined 3D world points.
[0111] The system that finds position and/or direction (e.g., orientation) of an object in motion includes at least one memory device and at least one processor coupled with the memory device, where the at least one processor is configured to perform at least the following operations. For example, a processor receives an image taken of an object in motion. Examples of the object include, but are not limited to, a baseball bat, a golf club, a cricket bat, another ball striking element, or an object that can be swung and whose position is desired to be known. The object has at least three markers. The markers can be of any predefined shape, which can be recognized in an image of the object. The markers may be attached to the object, for example, glued on or painted on, or by any other means placed on the object, e.g., before the image of the object in motion is captured. FIG. **9** shows an example of an object with three markers in some embodiments. The object **902** has three markers **904**, **906**, **908**. The actual length of the object or part of the length of the object where the three markers are positioned, referred to as L, is known; The distance between the markers is known. For example, the distance from first marker to second marker,

referred to as p.sub.12, is known; the actual distance from second marker to third marker, referred to as p.sub.23, is known.

[0112] In some embodiments, the image is captured by a camera such as a single camera or a mono camera that takes a snapshot or picture of the object in motion. The image is a two-dimensional (2D) image of the object in motion. In some embodiments, the image can be a frame among a sequence of frames (e.g., video frames) taken with a camera. The image is a two-dimensional (2D) image of the object in motion, e.g., captured by a single camera. For example, the camera can be coupled with the processor and configured to capture the image of the object in motion. In some embodiments, the camera has been calibrated without decomposing camera parameters into explicit extrinsic and explicit intrinsic camera parameters, for example, as described above with reference to camera calibration. In some embodiments, the camera may have been subjected to a factory calibration a described above with reference to camera calibration, where the extrinsic camera parameters are updated through an affine correction.

[0113] The processor detects in the image, at least three markers on the object. For instance, the processor may use any known image recognition techniques to recognize in the image the object in motion, the three markers on the object.

[0114] The processor determines image points of the three markers **904**, **906**, **908**. For explanation purposes, theses image points in 2D image are denoted as q (e.g., q.sub.1, q.sub.2, q.sub.3, or q.sub.n, where n is 1, 2, or 3 referring to n-th marker). FIG. **10** shows an example of an image taken of an object in motion with three markers using a mono camera, in some embodiments. An image point is represented by its 2D x-y coordinate on an image plane. Based on the image points (x-y coordinates of q.sub.1, q.sub.2, q.sub.3), the processor then determines the distance in the image between q.sub.1 and q.sub.2, referred to as q.sub.12, and the distance in the image between q.sub.2 and q.sub.3, referred to as q.sub.23.

[0115] The processor, based on the image points of the three markers, determines depths of the object at the image points, where the depths is determined relative to an image plane of the image, for example, as a reference point. FIG. **11** shows another perspective of the image captured of an object in motion in some embodiments. Image plane of the image is shown at **1102**. In some embodiments, this image plane aligns with a position of the camera **1104**. For purposes of explanation, depth is denoted as z, where z.sub.0 refers to depth at image plane, z.sub.01 refers to the depth of a first marker on the object relative to the image plane, z.sub.02 refers to the depth of a second marker on the object relative to the image plane, and z.sub.03 refers to the depth of a third marker on the object relative to the image plane. At image plane, depth z.sub.0=0. It is noted that the depth z, e.g., z.sub.01, z.sub.02, z.sub.03, is not the true depth; rather they refer to parameters that are mixed with arbitrary scale that the homography matrix H holds.

[0116] In some embodiment, the processor uses a homography matrix H and cross-ratio, to determine the depth of second marker, z.sub.02. Using z.sub.02, z.sub.01 and zos can be determined or derived. Then, based on the depths of the respective markers, z.sub.01, z.sub.02, and z.sub.03, 3D world points, p.sub.1, p.sub.2 and p3 can be determined. A derivation of the 3D world points is described in the following paragraphs, in some embodiments. For purposes of explanation, 3D world points are denoted as p, e.g., p.sub.1 refers to 3D real world point of a first marker (e.g., shown at **904**), p.sub.2 refers to 3D real world point of a second marker (e.g., shown at **906**), and p.sub.3 refers to 3D real world point of a third marker (e.g., shown at **908**); p.sub.0 refers to 3D real world point of the camera position, also referred to as p.sub.c.

[0117] The homography matrix H, in some embodiments, as described above, is a factory (or first stage) calibration homography matrix that has been updated or revised by performing a field (or second stage) calibration. As described with reference to camera calibration, the homography matrix, H comprises the plurality of extrinsic and intrinsic camera parameters.

[0118] Cross-ratio (λ) or λ.sub.2D is computed as follows:

[00011] $= \frac{q_{12}}{q_{23}}.$

That is, cross-ratio by 2D image is

[00012] $= \frac{q_{03}\, q_{12}}{q_{01}\, q_{23}}$ or $= \frac{q_{12}}{q_{23}}$

as 2D point {right arrow over (q)}.sub.0 is at infinity.

[0119] Cross-ratio by 3D (or λ.sub.3D) is computed as

[00013] $= \frac{p_{03} p_{12}}{p_{01} p_{23}} = \frac{p_{03}}{p_{01}} = \frac{z_{03}}{z_{01}}$, with $= \frac{p_{12}}{p_{23}}$;

[0120] Parameter “Beta” or β is a ratio defined above and is predefined (known). As described above, the actual distance between the three markers is known or predefined. That is the distance from first marker to second marker (p.sub.12), second marker to third marker (p.sub.23), first marker to third marker (p.sub.13) are known. In some embodiments, “Beta” or β is a ratio of distance between first and second markers and distance between second and third markers,

[00014] $= \frac{p_{12}}{p_{23}}$.

[0121] Projection: {right arrow over (p)}.sub.nH=z.sub.0n{right arrow over (q)}.sub.n; where H is a projection (homography) matrix, {right arrow over (p)}.sub.n are world points, {right arrow over (q)}.sub.n are image points, z.sub.n are unknown depths, subject to determination.

[0122] From {right arrow over (p)}.sub.nH=z.sub.0n{right arrow over (q)}.sub.n, one can obtain: {right arrow over (p)}.sub.n=z.sub.0n{right arrow over (q)}.sub.nH.sup.−1; {right arrow over (p)}.sub.n−{right arrow over (p)}.sub.c=z.sub.0n{right arrow over (q)}.sub.nH.sub.r.sup.−1; with H.sub.r being a rotational part of the homography matrix H. Here, p.sub.c refers to camera position, as described hereinabove.

[00015]

$$p_{13} = \sqrt{(\overset{.fwdarw.}{p}_3 - \overset{.fwdarw.}{p}_1)^2} = \sqrt{(z_{03}\overset{.fwdarw.}{q}_3 - z_{01}\overset{.fwdarw.}{q}_1)(H_r^T H_r)^{-1}(z_{03}\overset{.fwdarw.}{q}_3 - z_{01}\overset{.fwdarw.}{q}_1)^T};$$

[0123] Depths z.sub.0n are bound to each other by ratios λ and β as:

[00016] $z_{03} = -\frac{1+}{1+} z_{02}; z_{01} = \frac{1+}{1+} z_{02}; (z_{03}\overset{.fwdarw.}{q}_3 - z_{01}\overset{.fwdarw.}{q}_1) = \frac{1+}{1+} z_{02}(-\overset{.fwdarw.}{q}_3 - \overset{.fwdarw.}{q}_1);$

Then

[00017] $z_{02} = \frac{(1+\ )}{(1+\ )} \frac{p_{13}}{\sqrt{(-\overset{.fwdarw.}{q}_3 - \overset{.fwdarw.}{q}_1)(H_r^T H_r)^{-1}(-\overset{.fwdarw.}{q}_3 - \overset{.fwdarw.}{q}_1)^T}};$

[0124] With all known z.sub.0n, the processor determines 3D world points from {right arrow over (p)}.sub.n=z.sub.0n{right arrow over (q)}.sub.nH.sup.−1.

[0125] Using the image points q.sub.n, i.e., q.sub.1, q.sub.2, q.sub.3, and the depths z.sub.01, z.sub.02, z.sub.03, the processor determines 3D world points p.sub.n, i.e., p.sub.1, p.sub.2, p.sub.3, by the above formulation, {right arrow over (p)}.sub.n=z.sub.0n {right arrow over (q)}.sub.nH.sup.−1. The 3D world points represent a position of the object in a 3D real world coordinate.

[0126] In some embodiments, the three markers are distinct from one another. Based on the distinctive characteristics of the each of the markers and their respective 3D real world points, the processor determines the direction or orientation of the object in motion in the image. The processor determines the direction of the object in motion as a path along the 3D world points. For example, the direction or the orientation of the object in motion in 3D is along the {right arrow over (p)}.sub.3-{right arrow over (p)}.sub.1 3D vector.

[0127] FIG. **12** is a flow diagram illustrating a method of finding position of an object in motion in some embodiments. The method is performed by at least one computer processor, for example, as described with reference to FIGS. **9**, **10** and **11** above. At **1202**, the method includes receiving an image taken of an object in motion. As described hereinabove, the image may be captured by a mono camera that has been calibrated. At **1204**, the method includes detecting in the image, at least three markers on the object. At **1206**, the method includes determining image points of the three markers. At **1208**, the method includes, based on the image points of the three markers, determining depths of the object at the image points, the depths determined relative to an image plane of the image. At **1210**, the method includes, using the image points and the depths, determining three-dimensional (3D) world points representing a position of the object in a 3D real world coordinate. As described above, the three markers are distinct from one another, where the method further includes determining a direction of the object in motion based on the three markers. For example, as described above, the direction of the object is determined as a path along the 3D world points. As described above, determining of the depths of the object at the image points uses a homography matrix H and cross-ratio of the image points, where the homography matrix H maps 3D world points to two-dimensional (2D) image points. As described above, in some embodiments, a single camera coupled with the computer processor captures the image of the object in motion. As described above, in some embodiments, the single camera has been calibrated without decomposing camera parameters into explicit extrinsic and explicit intrinsic camera parameters.

[0128] A computer storage medium or media includes one or more storage devices that collectively include machine readable code corresponding to instructions and/or data for performing computer operations specified in a given computer storage medium claim. A “storage device” is any tangible device that can retain and store instructions for use by a computer processor. Without limitation, the computer readable storage medium may be an electronic storage medium, a magnetic storage medium, an optical storage medium, an electromagnetic storage medium, a semiconductor storage medium, a mechanical storage medium, or any suitable combination of the foregoing. Some known types of storage devices that include these mediums include, but are not limited to: diskette, hard disk, random access memory (RAM), read-only memory (ROM), erasable programmable read-only memory (EPROM or Flash memory), static random access memory (SRAM), compact disc read-only memory (CD-ROM), digital versatile disk (DVD), memory stick, floppy disk, mechanically encoded device (such as punch cards or pits/lands formed in a major surface of a disc) or any suitable combination of the foregoing. A computer readable storage medium, as that term is used in the present disclosure, is not to be construed as storage in the form of transitory signals per se, such as radio waves or other freely propagating electromagnetic waves, electromagnetic waves propagating through a waveguide, light pulses passing through a fiber optic cable, electrical signals communicated through a wire, and/or other transmission media. As will be understood by those of skill in the art, data is typically moved at some occasional points in time during normal operations of a storage device, such as during access, de-fragmentation or garbage collection, but this does not render the storage device as transitory because the data is not transitory while it is stored.

[0129] The specific embodiments of the present disclosure are described above, but various modifications to the specific embodiments are possibly implemented without departing from the scope of the present disclosure. However, it is apparent to a person of ordinary skill in the art to which the present disclosure pertains that various alterations and modifications are possible without departing from the nature and gist of the present disclosure. Therefore, the embodiments of the present disclosure are for describing the technical idea of the present disclosure, rather than limiting it, and do not impose any limitation on the scope of the technical idea of the present disclosure. Accordingly, the scope of the present disclosure should be defined by the following claims. All equivalent technical ideas should be interpreted to be included within the scope of the present disclosure.

Claims

1. A system comprising: at least one memory device; and at least one processor coupled with the memory device, the at least one processor configured to at least: receive an image taken of an object in motion; detect in the image, at least three markers on the object in motion; determine image points of the three markers; based on the image points of the three markers, determine depths of the object in motion at the image points, the depths determined relative to an image plane of the image; and using the image points and the depths, determine three-dimensional (3D) world points representing a position of the object in motion in a 3D real world coordinate.

2. The system of claim 1, wherein the three markers are distinct from one another, wherein the at least one processor is further configured to determine a direction of the object in motion based on the three markers.

3. The system of claim 2, wherein the direction of the object in motion is determined as a path along the 3D world points.

4. The system of claim 1, wherein the at least one processor is configured to use a homography matrix H and cross-ratio of the image points to determine the depths, the homography matrix H mapping 3D world points to two-dimensional (2D) image points.

5. The system of claim 4, wherein the image is a 2D image of the object in motion captured by a single camera.

6. The system of claim 5, wherein the single camera has been calibrated without decomposing camera parameters into explicit extrinsic and explicit intrinsic camera parameters.
7. The system of claim 1, further including a single camera coupled with the at least one processor, and configured to capture the image of the object in motion.
8. The system of claim 7, wherein the single camera has been calibrated without decomposing camera parameters into explicit extrinsic and explicit intrinsic camera parameters.
9. A method performed by a least one processor, the method comprising: receiving an image taken of an object in motion; detecting in the image, at least three markers on the object in motion; determining image points of the three markers; based on the image points of the three markers, determining depths of the object in motion at the image points, the depths determined relative to an image plane of the image; using the image points and the depths, determining three-dimensional (3D) world points representing a position of the object in motion in a 3D real world coordinate.
10. The method of claim 9, wherein the three markers are distinct from one another, wherein the method further includes determining a direction of the object in motion based on the three markers.
11. The method of claim 10, wherein the direction of the object in motion is determined as a path along the 3D world points.
12. The method of claim 9, wherein the determining the depths of the object in motion at the image points uses a homography matrix H and cross-ratio of the image points, the homography matrix H mapping 3D world points to two-dimensional (2D) image points.
13. The method of claim 9, further including capturing by a single camera coupled with the at least one processor, the image of the object in motion.
14. The method of claim 13, wherein the single camera has been calibrated without decomposing camera parameters into explicit extrinsic and explicit intrinsic camera parameters.
15. A computer readable storage medium storing a program of instructions executable by a machine to perform a method of: receiving an image taken of an object in motion; detecting in the image, at least three markers on the object in motion; determining image points of the three markers; based on the image points of the three markers, determining depths of the object in motion at the image points, the depths determined relative to an image plane of the image; using the image points and the depths, determining three-dimensional (3D) world points representing a position of the object in motion in a 3D real world coordinate.
16. The computer readable storage medium of claim 15, wherein the three markers are distinct from one another, wherein the method further includes determining a direction of the object in motion based on the three markers.
17. The computer readable storage medium of claim 16, wherein the direction of the object in motion is determined as a path along the 3D world points.
18. The computer readable storage medium of claim 15, wherein the determining the depths of the object in motion at the image points uses a homography matrix H and cross-ratio of the image points, the homography matrix H mapping 3D world points to two-dimensional (2D) image points.
19. The computer readable storage medium of claim 15, wherein the image is a 2D image of the object in motion captured by a single camera.
20. The computer readable storage medium of claim 19, wherein the single camera has been calibrated without decomposing camera parameters into explicit extrinsic and explicit intrinsic camera parameters.
